

Data Structures



Lecture 14

Heaps

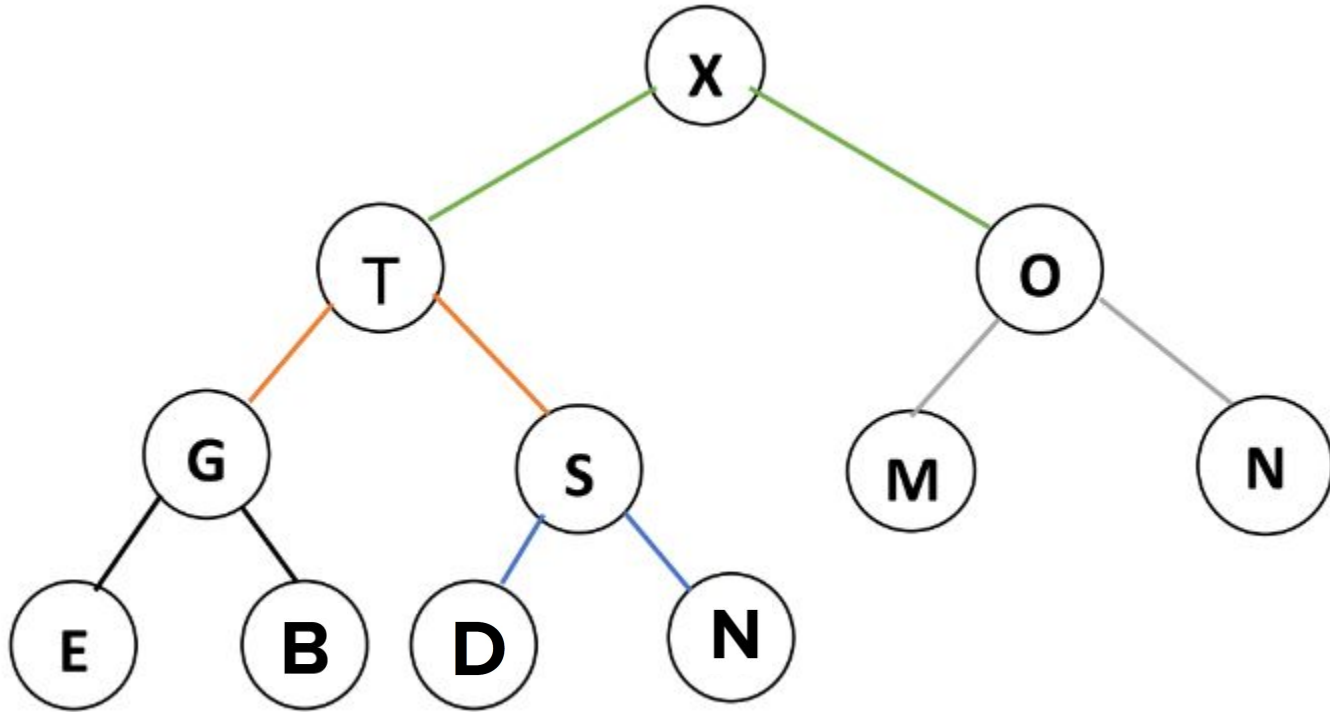
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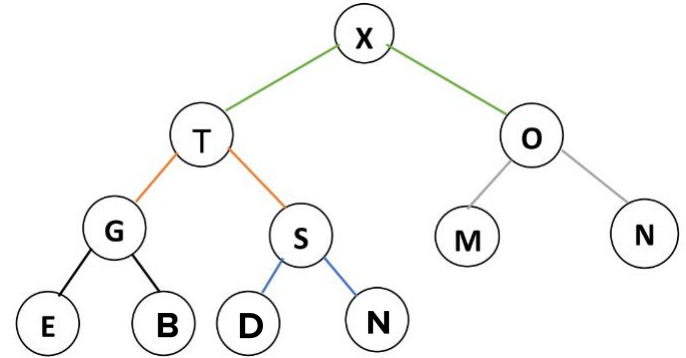
Heap - Special Binary Tree



Heap - Properties

→ Complete

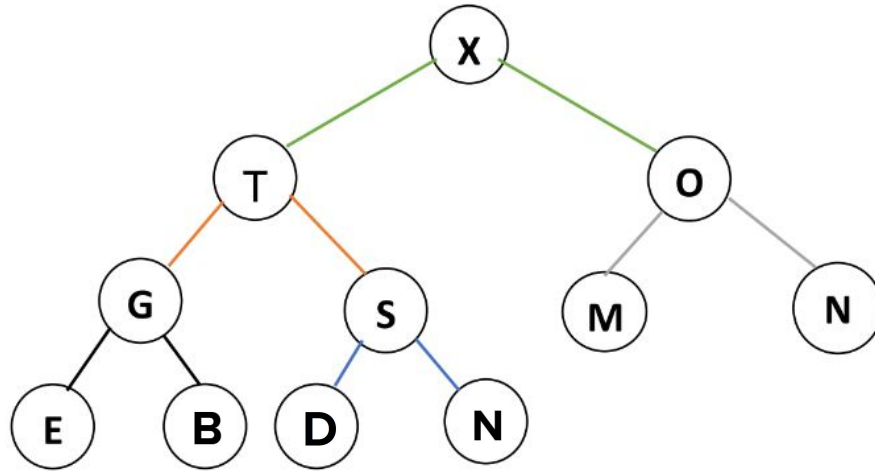
→ Satisfy **Heap Property**



→ Value of Parent $\geq$ Value of Children **MaxHeap**

→ Value of Parent $\leq$ Value of Children **MinHeap**

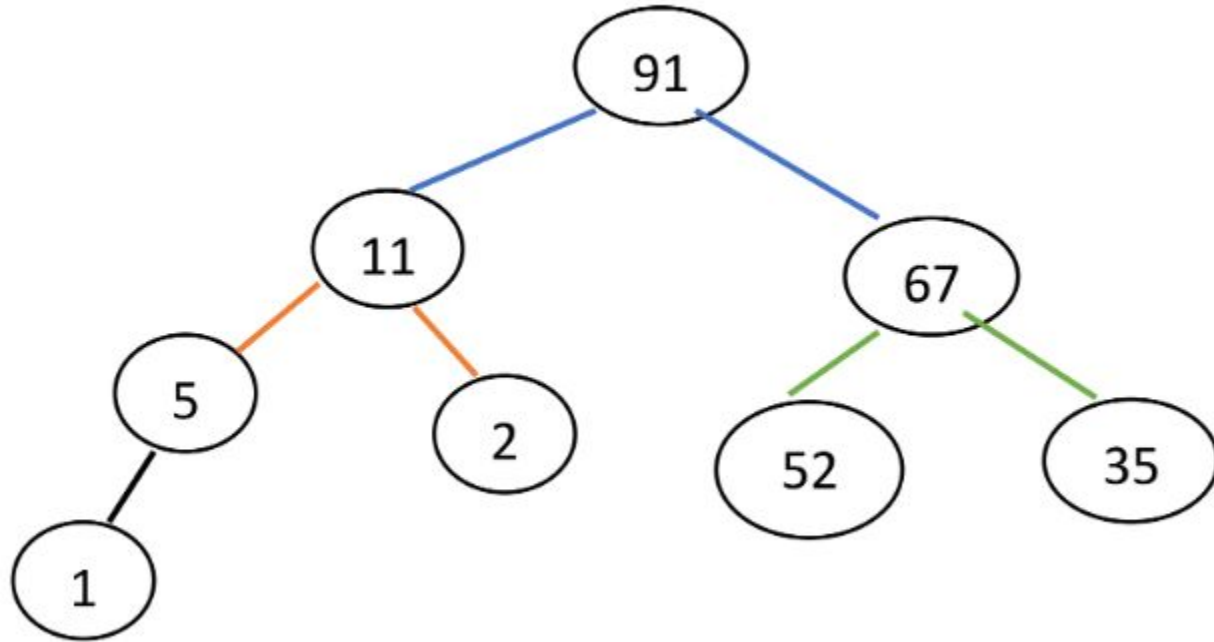
Heap - Representation



1	2	3	4	5	6	7	8	9	10	11
X	T	O	G	S	M	N	E	B	D	N

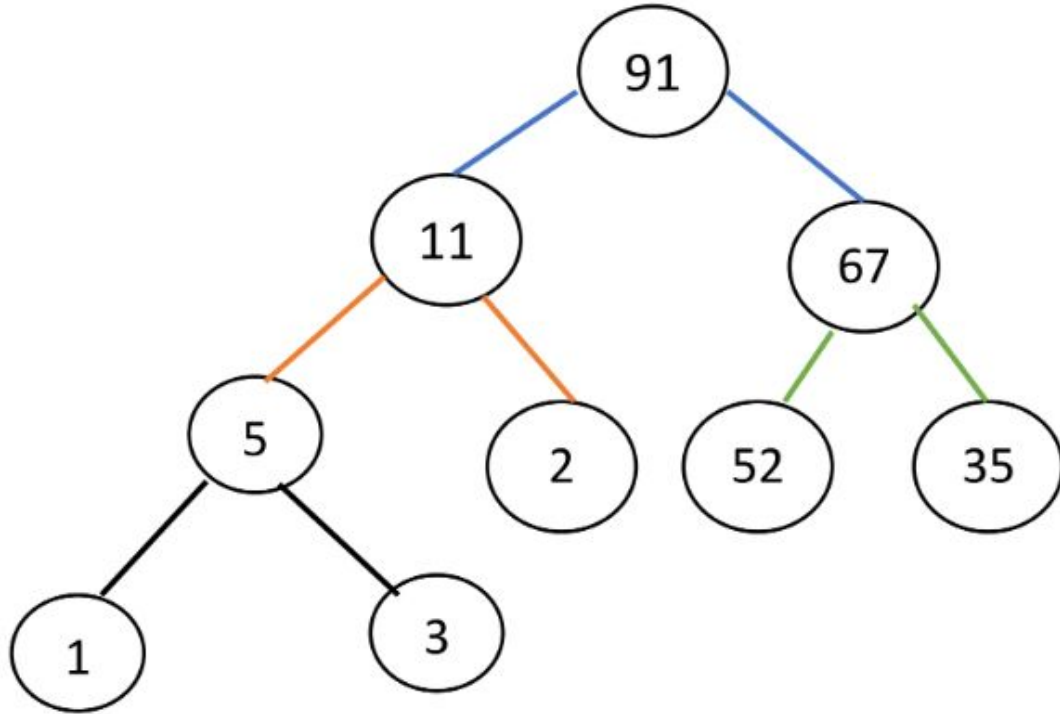
Heap - Insert

Insert 3



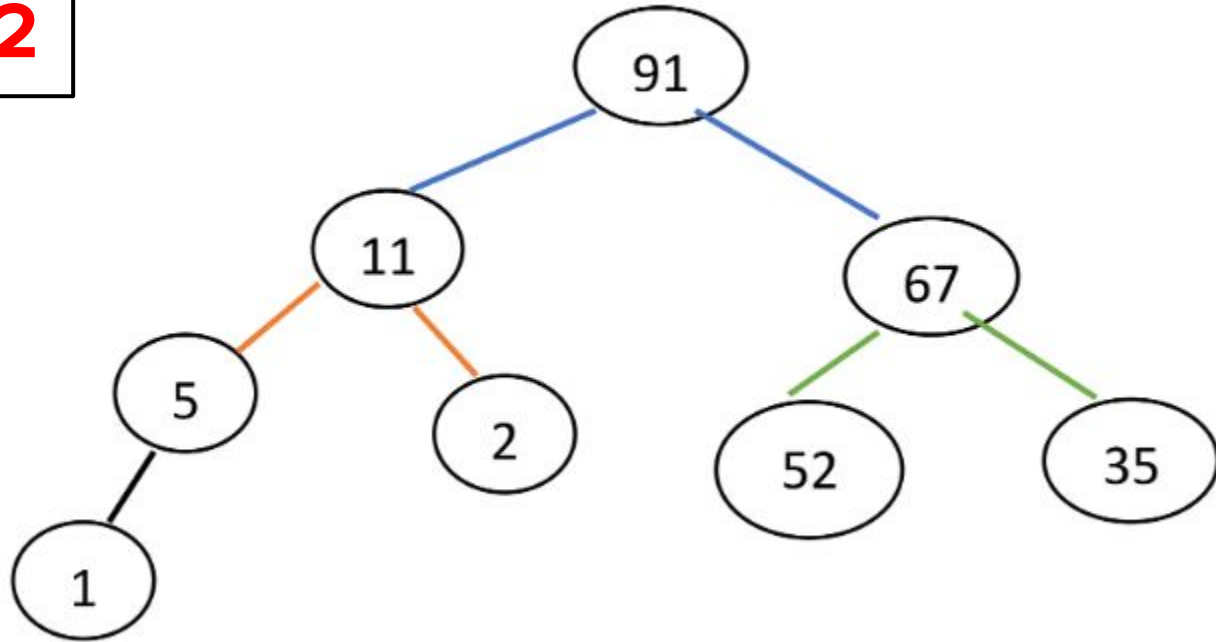
Heap - Insert

Insert 3



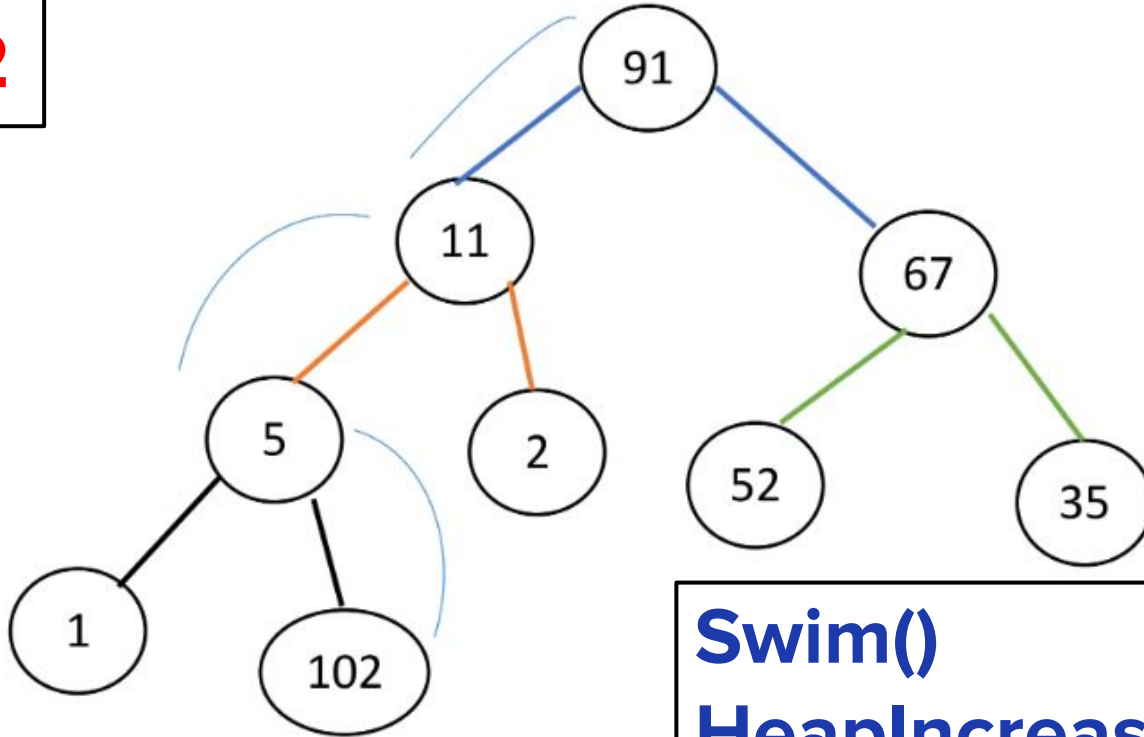
Heap - Insert

Insert 102



Heap - Insert

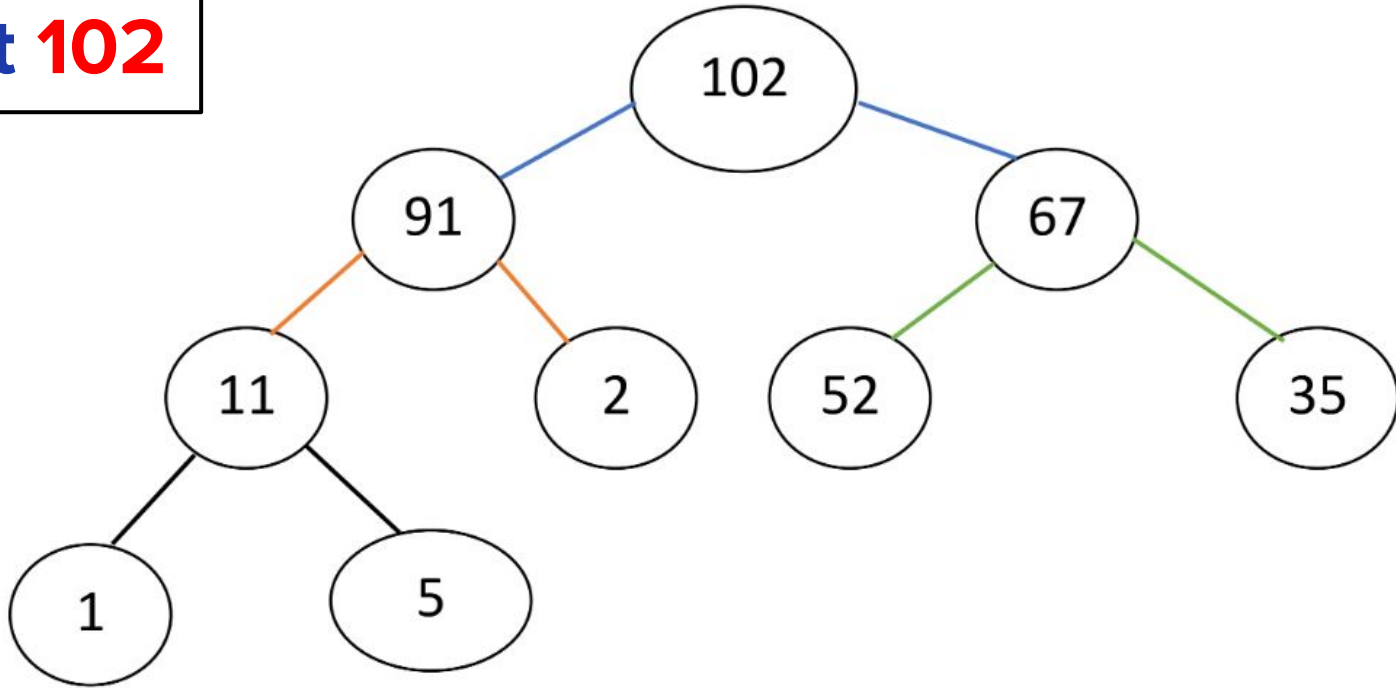
Insert 102



Swim()
HeapIncreaseKey()

Heap - Insert

Insert 102



Heap - Insert (Pseudocode)

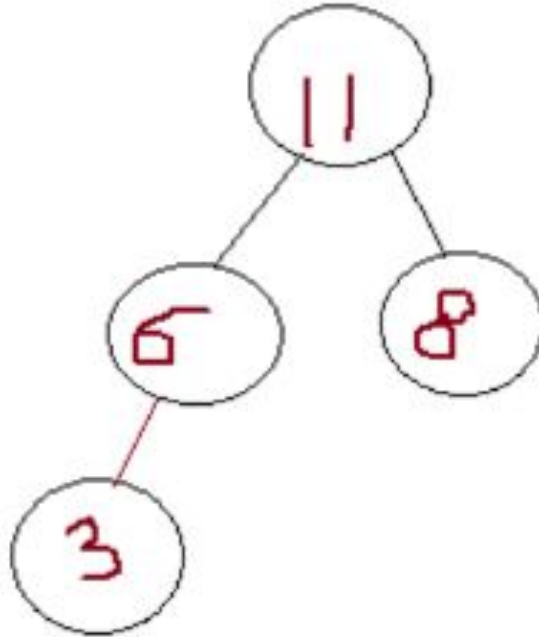
```
insert (H, key){  
    size(H) = size(H) + 1;  
    H[size] = key;  
    swim (H, size);  
}
```

Heap - Swim (Pseudocode)

```
swim(H, index){  
    if (index <= 1){  
        return;  
    }else{  
        parent = H[index/2];  
        if (parent > H[index]){  
            return;  
        }else{  
            exchange parent with H[index]  
            swim(H, parent);  
        }  
    }  
}
```

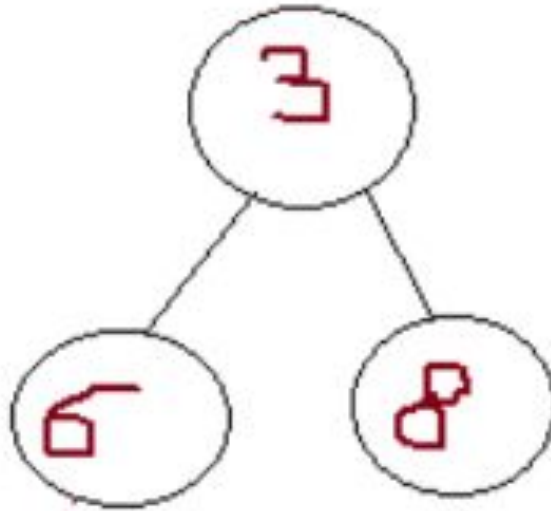
Heap - Delete

Delete Root



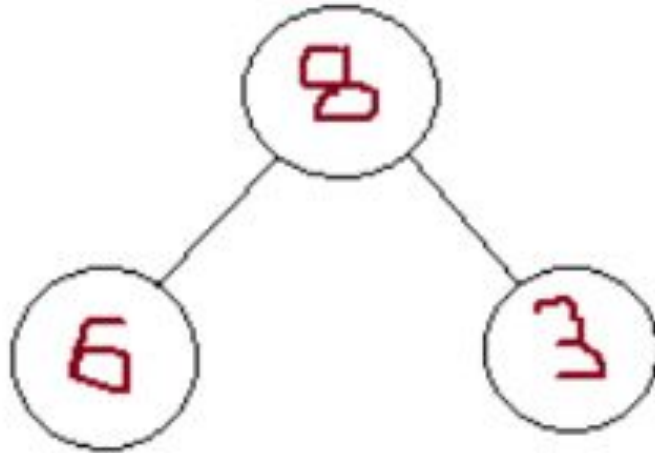
Heap - Delete

Delete Root



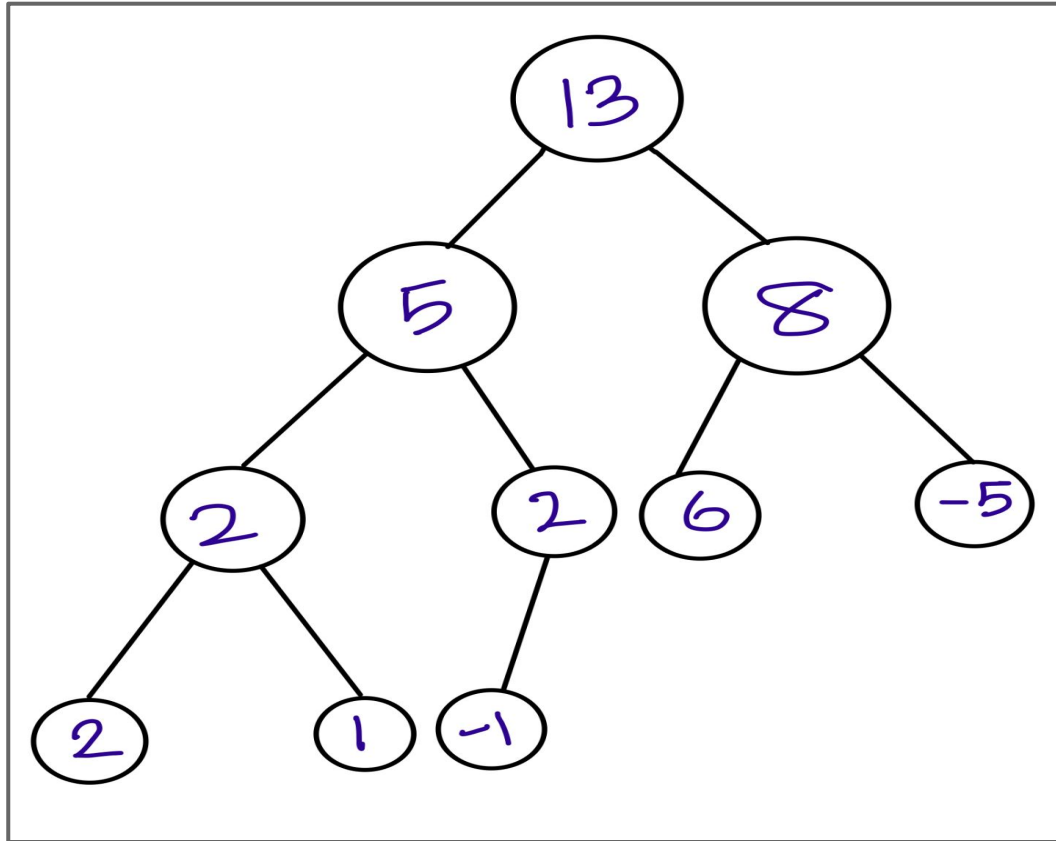
Heap - Delete

Delete Root



Sink()
MaxHeapify()

Heap - Delete



Heap - Delete (Pseudocode)

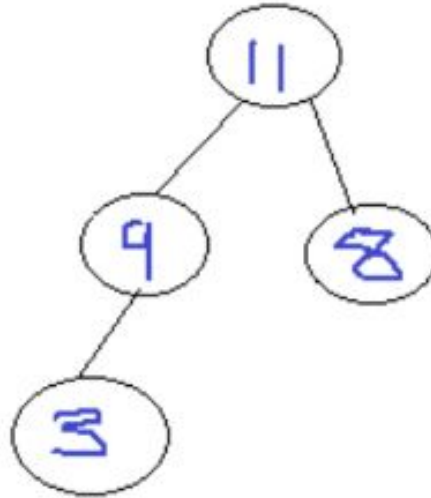
```
delete(H){  
    if (size(H)==0){  
        return;  
    }else{  
        exchange H[1] with H[size]  
        size --;  
        maxHeapify(H, 1)  
    }  
}
```


Heap - Sink (Pseudocode)

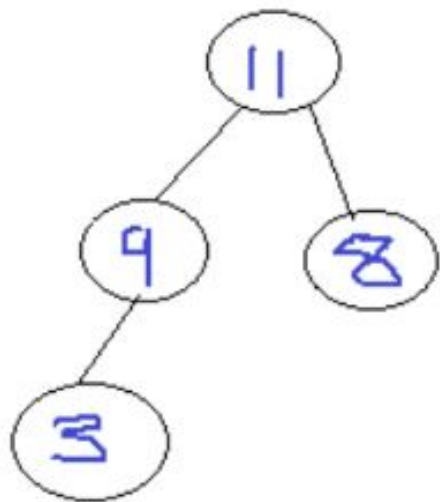
```
maxHeapify(H, index){
    if (size(H) == 0){
        return;
    } else {
        left = 2*index;
        right = 2*index+1;
        if (left <= size && right <= size){
            exchange H[1] with Max (H[left], H[right]);
            maxHeapify(Max (left, right));
        } else {
            if (left <= size && right > size){
                exchange H[1] with (H[left]);
            }
        }
    }
}
```

Heap Sort

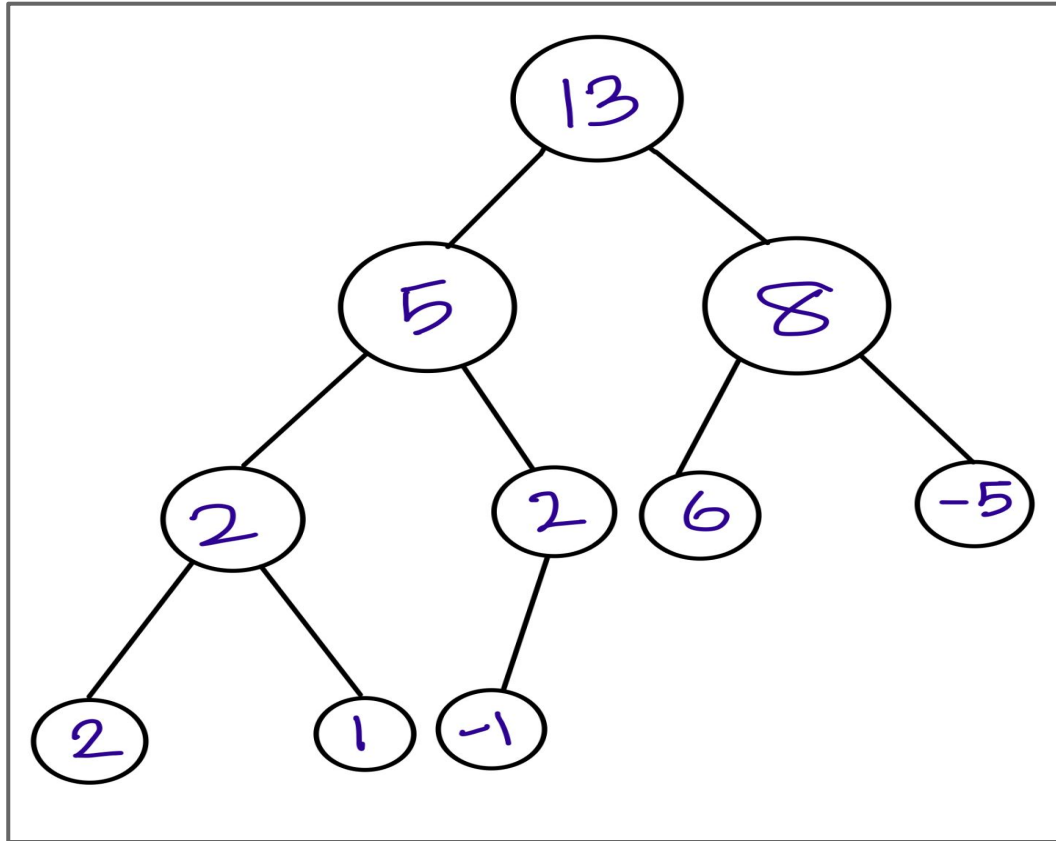
Delete All Nodes



Heap Sort



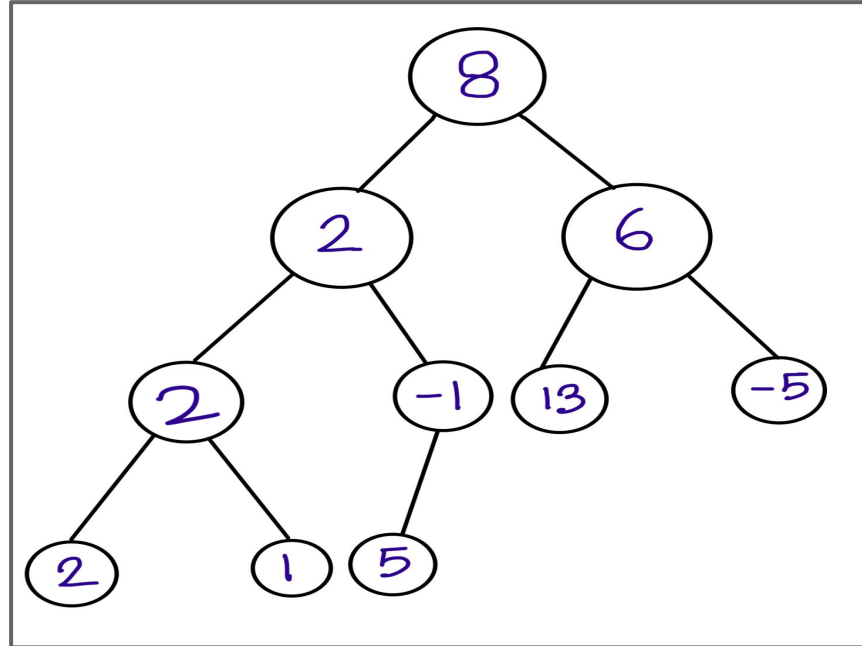
Heap Sort



Heap - Build Heap

N	8	2	6	2	-1	13	-5	2	1	5
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Tree

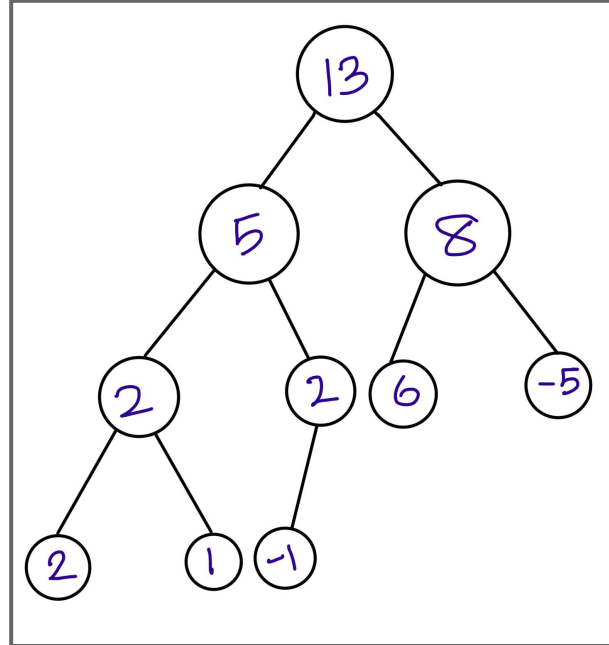


Heap - Build Heap

N	8	2	6	2	-1	13	-5	2	1	5
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```
for all nodes i = 1 to n{  
    swim (H, i);  
}
```

Heap



Heap - Build Heap

N	13	5	8	2	2	6	-5	2	1	-1
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```
for all nodes i = 1 to n{  
    swim (H, i);  
}
```

Heap

